

CSA5114-CRYPTOGRAPHY AND NETWORK SECURITY

1. Write a C program for Caesar cipher involves replacing each letter of the alphabet with the letter standing k places further down the alphabet, for k in the range 1 through 25.

Output

```
Enter the text: cryptography
Enter the key (1-25): 3
Encrypted Text: fubswrjudskb

=== Code Execution Successful ===
```

2. Write a C program for monoalphabetic substitution cipher maps a plaintext alphabet to a ciphertext alphabet, so that each letter of the plaintext alphabet maps to a single unique letter of the ciphertext alphabet.

Output

```
Enter the text: ryptography
Encrypted Text: knhzgukqhln

=== Code Execution Successful ===
```

3. Write a C program for Playfair algorithm is based on the use of a 5 X 5 matrix of letters constructed using a keyword. Plaintext is encrypted two letters at a time using this matrix.

Output

```
Enter two uppercase letters: HI
Encrypted Text: BF

=== Code Execution Successful ===
```

4. Write a C program for polyalphabetic substitution cipher uses a separate monoalphabetic substitution cipher for each successive letter of plaintext, depending on a key.

Output

```
Enter the plaintext: cryptography
Enter the key: KEY
Encrypted Text: mvwzxm qvyzlw

=== Code Execution Successful ===
```

5. Write a C program for generalization of the Caesar cipher, known as the affine Caesar cipher, has the following form: For each plaintext letter p , substitute the ciphertext letter C : $C = E([a, b], p) = (ap + b) \bmod 26$. A basic requirement of any encryption algorithm is that it be one-to-one. That is, if $p \neq q$, then $E(k, p) \neq E(k, q)$. Otherwise, decryption is impossible, because more than one plaintext character maps into the same ciphertext character. The affine Caesar cipher is not one-to-one for all values of a . For example, for $a = 2$ and $b = 3$, then $E([a, b], 0) = E([a, b], 13) = 3$. a. Are there any limitations on the value of b ? b. Determine which values of a are not allowed.

Output

```
Enter the plaintext: CRYPTOGRAPHY
Enter the value of a: 5
Enter the value of b: 8
Encrypted Text: SPYFZAMPIFRY
```

```
=== Code Execution Successful ===
```